Hasan Cem Gençoğlu
150719565

On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

1-)

$$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

a) $R_1 \cdot -3\alpha + R_2 \Rightarrow$
$-2R_1 + R_3 \rightarrow$
$R_4 - R_1 \rightarrow$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5-3\alpha^2 & 3\alpha\gamma-9\gamma & 1-3\alpha^2 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix} \qquad \alpha = 0$$

$$\begin{bmatrix} 1 & 0 & -\gamma & 0 \\ 0 & 5 & -9\gamma & 1 \\ 0 & \beta & 3+2\gamma & 0 \\ 0 & 14 & \gamma-6 & 2 \end{bmatrix}$$

$R_1 \cdot R_3 \Rightarrow R_3$
$R_1 \cdot R_4 \Rightarrow R_4$

$$\begin{bmatrix} 1 & 0 & -\gamma & 0 \\ 0 & 5 & -9\gamma & 1 \\ 0 & 0 & -3\gamma-2\gamma^2 & 0 \\ 0 & 0 & -\gamma^2+6\gamma & 0 \end{bmatrix}$$

$-2R_3 + R_4 \Rightarrow R_4$

$$\begin{bmatrix} 1 & 0 & -\gamma & 0 \\ & & & \end{bmatrix}$$

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3-/ a) v is a unit vector in $\mathbb{R}^n$. So $v^T v = 1$

$H^2 = (I - 2vv^T)^2 = I^2 - 4vv^T + 4vvv^Tv^T$ since $H^2 = I$ $v^Tv = 1$

$H^T = H \Rightarrow H^T H = 1$ Because H is an orthogonal matrix.

b)

c) $Hv = (I - 2vv^T)v = v - 2vv^Tv = v - 2v = -v$, $\lambda = -1$.

d) For $v^Tv = 0$ $Hv = (I - 2vv^T)v = v - 2vv^Tv - 2v$

So λ is 1. Orthogonal complement has dimension $(n-1)$.

Adding in the eigenvector v, we find that H is diagonalizable.

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